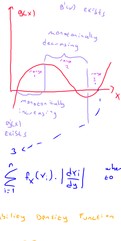
The method of transformations

(read topic in probability book)



Theorem of method of transformations

- Let X continuous r.v. with R_X
- suppose we can partition R_X into disjoint sets such that $g(x)$ is strictly monotone and differentiable on each partition
- Then we can write $f_Y(y) = \sum_{i=1}^{n} \frac{f_X(x_i)}{|g'(x_i)|} = \sum_{i=1}^{n} f_X(x_i) \cdot \left|\frac{dx_i}{dy}\right|$ where x_i are real solutions to $g(x)=y$

n = # partitions

X is continuous r.v. with Probability Density Function

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} \quad \text{for all } x \in \mathbb{R}$$

Let $Y = X^2$ find the $f_Y(y)$

solution: $g(x) = x^2$

x_1 is solution to $g(x)=y$ in partition 1

$x_1 = g_1^{-1}(y) = -\sqrt{y}$

x_2 is solution to $g(x)=y$ in partition 2

$x_2 = g_2^{-1}(y) = \sqrt{y}$

$g(x) = x^2 \quad g'(x) = 2x$

$$f_Y(y) = \sum_{i=1}^{2} \frac{f_X(x_i)}{|g'(x_i)|} = \frac{1}{|-2\sqrt{y}|} \frac{e^{\frac{-y}{2}}}{\sqrt{2\pi}} + \frac{1}{|2\sqrt{y}|} \frac{e^{\frac{-y}{2}}}{\sqrt{2\pi}} \Rightarrow \frac{1}{\sqrt{2\pi}} \frac{1}{|2\sqrt{y}|} e^{\frac{-y}{2}} + \frac{1}{\sqrt{2\pi}} \frac{1}{|2\sqrt{y}|} e^{\frac{-y}{2}} = \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{\frac{-y}{2}} + \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{\frac{-y}{2}} = \frac{e^{\frac{-y}{2}}}{\sqrt{2\pi}\sqrt{y}}, \text{ for } y \in (0,\infty)$$

R_Y

Special C.R.V. distribution

Uniform distribution

definition: A C.R.V. X has uniform distribution over $[a,b]$ if $f_X(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$

we write $X \sim \text{Uniform}(a,b)$

Graph

$$EX = \frac{a+b}{2} \qquad Var(X) = \frac{(b-a)^2}{12}$$

Exponential dist

used for time between events

definition: A C.R.V. X has exponential dist with $\lambda > 0$, then $X \sim \text{Exponential}(\lambda)$

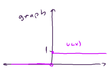
$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \ge 0 \\ 0 & \text{otherwise} \end{cases}$$

graph

Unit step function

$$u(x) = \begin{cases} 1 & x \ge 0 \\ 0 & \text{otherwise} \end{cases}$$

graph



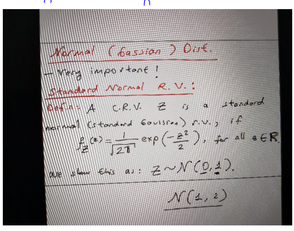
Next: exponential distribution using Unit Step function

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \ge 0 \\ 0 & \text{otherwise} \end{cases}$$

$$= \lambda e^{-\lambda x} u(x)$$

EX and Var(X)

$EX = \frac{1}{\lambda} \qquad Var(X) = \frac{1}{\lambda^2}$ } no need to use integration by parts



Graph

symmetric about 0

$EZ = E\,Z^3 = E\,Z^5 \ldots = 0$

$Var(Z) = 1 \qquad \sigma_Z = 1$

CDF: No closed form solution

$$F_Z(z) = \int_{-\infty}^{z} f_Z(u)\,du = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{\frac{-u^2}{2}} du$$

no solution for this

we do have numerical approximations

$\Phi\left(\frac{\theta-(-\gamma)}{2}\right)$

$\Phi\left(\frac{-\gamma-(-\gamma)}{2}\right) = \Phi$

$x < -\gamma$

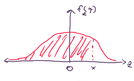
Φ is the symbol for CDF of $N(0,1)$

$F_Z(z) = \Phi(z)$ recall $F_Z(z) = P(Z \le z)$

Obtaining $\Phi(z)$: Ex: $\Phi(1.6)$

① Look at a Φ table. ② Favorite Programming language tool.

Φ in PDF graph



$$\Phi(x) = F_X(x) = P(Z \le x) = \int_{-\infty}^{x} f_Z(u)\,du = \text{area}$$

Φ graph

properties of Φ

$\Phi(-x) = 1 - \Phi(x)$ for all $x \in \mathbb{R}$

Normal random variables

Z: standard normal variables

$X = \sigma Z + M$, where $\sigma > 0$

$EX = M \qquad Var(X) = \sigma^2$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-M)^2}{2\sigma^2}\right)$$

X: normal random variable

$X \sim N(M, \sigma^2)$

$$F_X(x) = P(X \le x) = P(Z \le \tfrac{x-M}{\sigma}) = \Phi\left(\tfrac{x-M}{\sigma}\right)$$

no closed form

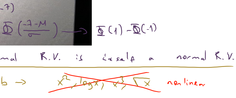
$X = \sigma Z + M$

$Z = \frac{X-M}{\sigma}$

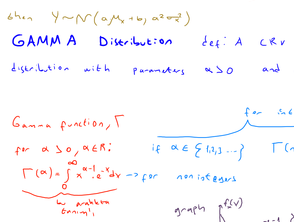
$P(a < X \le b) = F_X(b) - F_X(a)$

$= \Phi\left(\frac{b-M}{\sigma}\right) - \Phi\left(\frac{a-M}{\sigma}\right)$

Graph Normal dist.



Variance ≈ spread



Linear transformation of Normal R.V. is itself a normal R.V.

If $X \sim N(M_X, \sigma_X^2)$, $Y = aX + b \rightarrow$ ~~$X = \log(Y) - b$~~ ~~Y~~ nonlinear

then $Y \sim N(aM_X + b, a^2\sigma_X^2)$

GAMMA Distribution

def: A CRV X has a gamma distribution with parameters $\alpha > 0$ and $\lambda > 0$ if $f_X(x) = \begin{cases} \frac{\lambda^\alpha \cdot x^{\alpha-1} \cdot e^{-\lambda x}}{\Gamma(\alpha)} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$

Gamma function, Γ

for $\alpha > 0$, $\alpha \in \mathbb{R}$:

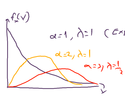
$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\,dx$ → for non integers

to problem tuning!

for integers

if $\alpha \in \{1, 2, \ldots\}$ $\Gamma(\alpha) = (\alpha-1)!$

graph



GAMMA and Exponential dists

1-) If $\alpha = 1$, gamma becomes Exponential

GAMMA$(1,\lambda)$ = Exponential(λ)

2-) If you sum n independent Exponential r.v's you get a gamma(n,λ) r.v.

EX and variance

$EX = \frac{\alpha}{\lambda} \qquad Var(X) = \frac{\alpha}{\lambda^2}$